

ANNOUNCING GLOBAL OPTIMISATION

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► Suppose we wish to minimize $f(x)$ possibly non-convex function.

► Consider the regularised problem for $\lambda > 0$:

$$\arg \min_x f(x) + \lambda g(x)$$

$$= \arg \max_x \exp(-\beta f(x) - g(x))$$

$$= \arg \min_x \beta f(x) + g(x), \quad \beta = 1/\lambda$$

$$= \arg \max_x \pi_\beta(x)$$

$$\pi_{\beta}(x) \propto \exp(-\beta f(x)) - g(x)$$



Where for $\beta \geq 0$ define



Mean view the linear path as a single MAPE estimate for global optimization problem

► The regulariser is encoded by the reference

▶ The annealing parameter is the level of regularisation

▶ objective is encoded by the weight

ANNEALING AS GLOBAL OPTIMISATION

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- ▶ Suppose we wish to minimize $f(x)$ possibly non-convex function.
- ▶ Consider the regularised problem for $\lambda > 0$:

$$\begin{aligned}\arg \min_x f(x) + \lambda g(x) &= \arg \min_x \beta f(x) + g(x), \quad \beta = 1/\lambda \\ &= \arg \max_x \exp(-\beta f(x) - g(x)) \\ &= \arg \max_x \pi_\beta(x)\end{aligned}$$

- ▶ Where for $\beta \geq 0$ define

$$\pi_\beta(x) \propto \exp(-\beta f(x) - g(x))$$

- ▶ We can view the linear path as encoding MAP estimate for global optimization problem
 - ▶ The regulariser is encoded by the reference
 - ▶ The annealing parameter encodes the level of regularisation
 - ▶ Objective is encoded by the weight

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